

What is Networking?

A complete, beginner-friendly guide with every task, question, explanation and flag.

Learning Path: Pre-Security / Network Fundamentals

Difficulty: Easy | Topics: Networks, Internet, IP, MAC, ICMP

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About this walkthrough

This document walks through every task of the TryHackMe room **What is Networking?**, part of the Pre-Security path. Each task includes the concept being taught, the exact questions asked, a short explanation of how to reach the answer, and the answer or flag itself.

What you will learn

- What a network is, and how the Internet is built from smaller networks.
- How devices identify themselves with IP and MAC addresses.
- How to test connectivity with the ping command over ICMP.
- How MAC spoofing works, demonstrated in an interactive lab.

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Before you start

Try the tasks yourself before reading the answers - that is where the real learning happens. Screenshots and exact UI wording in this room can change over time, so if a button label differs, look for the closest equivalent. All flags are shown for educational purposes after you have attempted the task.

TASK 1

What is Networking?

In simple terms, a network is just **things that are connected**. Your group of friends is a network, connected by shared interests. A city's public transport system, the national power grid, and the postal service are all networks too.

In computing, the same idea is applied to technological devices. A network can be formed by anywhere from **two devices to billions** - laptops, phones, security cameras, traffic lights, even farm sensors. Because networks are woven into everyday life, networking is a core concept in cyber security.

Think about it

- A network is any set of connected devices that can communicate.
- Networks come in all shapes and sizes - home LANs, corporate networks, and the global Internet.
- Everything you do online (weather data, streaming, banking) relies on networks.

Q1. What is the key term for devices that are connected?

ANSWER

Network

Key takeaway

If two or more devices can communicate, they form a network. The term for those connected devices, and the concept itself, is simply **network**.

TASK 2

What is the Internet?

The **Internet** is one giant network that is made up of many, many smaller networks connected together. Imagine Alice makes new friends (Zayn and Toby) who speak a different language. Alice is the only one who can speak both languages, so she acts as the messenger - and in doing so, a new, larger network is formed.

A short history

The first iteration of the Internet was the **ARPANET** project in the late 1960s, funded by the United States Department of Defense - the first documented network in action. However, the Internet as we know it was invented in **1989** by **Tim Berners-Lee** with the creation of the World Wide Web (WWW). From that point the Internet became the repository for storing and sharing information that it is today.

Private vs. public networks

Network type	What it is	Example
Private network	A small, self-contained network made of connected devices	Your home Wi-Fi LAN
Public network	The network that connects all the small private networks together	The Internet

Q1. Who invented the World Wide Web?

ANSWER

Tim Berners-Lee

Key takeaway

The Internet = many private networks joined by public networks. Devices use labels (IP and MAC addresses) to identify themselves on these networks - the topic of the next task.

TASK 3

Identifying Devices on a Network

To communicate in an orderly way, devices must be both **identifying** and **identifiable**. Humans are similar in that we have two ways of being identified: our **name** and our **fingerprints**. Devices mirror this with two identifiers:

- **IP Address** - changeable, like a name.
- **MAC Address** - permanent, like a fingerprint or serial number.

IP Addresses

An IP (Internet Protocol) address identifies a host on a network for a period of time. It is a set of numbers split into **four sections called octets**. IP addresses can change from device to device, but the same address cannot be active more than once within the same network.

```
192 . 168 . 1 . 77
----
Octet 1 Octet 2 Octet 3 Octet 4 -> 4 octets, each 0-255
```

IP addresses follow standards called **protocols**. Depending on where a device sits, it uses a public or a private IP address:

Device name	IP address	IP address type
DESKTOP-KJE57FD	192.168.1.77	Private
DESKTOP-KJE57FD	86.157.52.21	Public
CMNatic-PC	192.168.1.74	Private
CMNatic-PC	86.157.52.21	Public

Notice both devices share the same **public** IP on the Internet, but have unique **private** IPs on the local network. Public IPs are assigned by your **ISP** for a monthly fee.

IPv4 vs. IPv6

IPv4 uses a 32-bit scheme, giving roughly 4.29 billion addresses (2^{32}). With billions of new devices coming online each year, we ran out. **IPv6** supports 2^{128} addresses (340 trillion-plus) and is more efficient:

Version	Bit width	Total addresses	Example address
IPv4	32-bit	2^{32} (~4.29 billion)	192.168.1.77
IPv6	128-bit	2^{128} (340 trillion+)	2001:db8::ff00:42:8329

MAC Addresses

Every device has a physical network interface with a unique address assigned at the factory: the **MAC (Media Access Control)** address. It is a **twelve-character hexadecimal** number written in six pairs separated by colons, e.g. a4:c3:f0:85:ac:2d.

```
a4:c3:f0 : 85:ac:2d
|           |
first 6     last 6 characters
= the      = the device's
MANUFACTURER UNIQUE number (serial)
```

Unlike IP addresses, MAC addresses can be faked - a process called **spoofing**. A device pretends to be another by using its MAC address. This often breaks poorly designed security that trusts any device whose MAC address it recognises.

Practical - the hotel Wi-Fi lab

Deploy the interactive lab with the **View Site** button. Bob's packets (blue) are being thrown into the bin because he has not paid for Wi-Fi, while Alice's packets (green) pass through. The router filters devices by MAC address. **Copy Alice's MAC address and paste it into Bob's MAC field** (spoofing Bob as Alice), then click **Request Site** - the packet now passes and the flag is revealed.

Q1. What does the term "IP" stand for?

ANSWER

Internet Protocol

Q2. What is each section of an IP address called?

ANSWER

Octet

Q3. How many sections (in digits) does an IP address have?

ANSWER

4

Q4. What does the term "MAC" stand for?

ANSWER

Media Access Control

Q5. Deploy the interactive lab, spoof your MAC address to access the site. What is the flag?

ANSWER

THM{YOU_GOT_ON_TRYHACKME}

TASK 4

Ping (ICMP)

Ping is one of the most fundamental network tools. It uses **ICMP (Internet Control Message Protocol)** packets to determine the performance of a connection between devices - whether the connection exists and how reliable it is.

Ping measures the time taken for ICMP packets to travel between devices using an ICMP **echo** packet and an ICMP **echo reply**. It is installed by default on Linux, Windows and macOS.

Syntax

```
ping <IP address or website URL>

> ping 192.168.1.254
PING 192.168.1.254: 56 data bytes
64 bytes from 192.168.1.254: icmp_seq=0 ttl=63 time=5.3 ms
...
5 packets transmitted, 5 packets received, 0.0% packet loss
```

In the deployable lab, you ping **8.8.8.8** (Google's public DNS server) through a simulated terminal. The simulation sends the command `ping -c 4 8.8.8.8`, where the **-c** flag limits the number of packets to **4**. Without **-c**, ping runs forever until you press Ctrl+C.

Q1. What protocol does ping use?

ANSWER

ICMP

Q2. What is the syntax to ping 10.10.10.10?

ANSWER

ping 10.10.10.10

Q3. What flag do you get when you ping 8.8.8.8?

ANSWER

THM{I_PINGED_THE_SERVER}

TASK 5

Continue Your Learning - Intro to LAN

This final task simply points you to the next room in the Pre-Security path: **Intro to LAN**. Here you will build on everything in this room by learning about local area networks, topologies, switching and more.

Q1. Join the "Intro to LAN" room.

ANSWER

No answer needed - just click the green "Question Done" button.

Conclusion

This room lays the foundation for everything else in networking and security. You learned that a **network** is simply connected devices, that the **Internet** is a giant network of smaller networks, how devices identify themselves with **IP** (changeable) and **MAC** (permanent) addresses, and how to test connectivity with **ping** over ICMP.

Answer key at a glance

Task	Question	Answer
1	Key term for connected devices	Network
2	Inventor of the World Wide Web	Tim Berners-Lee
3	What "IP" stands for	Internet Protocol
3	Name for each section of an IP address	Octet
3	How many sections an IP address has	4
3	What "MAC" stands for	Media Access Control
3	Flag from the MAC spoofing lab	THM{YOU_GOT_ON_TRYHACKME}
4	Protocol used by ping	ICMP
4	Syntax to ping 10.10.10.10	ping 10.10.10.10
4	Flag from pinging 8.8.8.8	THM{I_PINGED_THE_SERVER}
5	Join the Intro to LAN room	No answer needed

Next steps

Continue with **Intro to LAN**, then **OSI Model** and **Packets & Frames** to complete the Network Fundamentals module. Practise the ping command on your own network and inspect your IP and MAC addresses with `ipconfig /all` (Windows) or `ip addr` (Linux).